



Clim1001quiz2 - practice

Introduction to Climate Change (University of New South Wales)



Scan to open on Studocu

What makes the ocean's structure different from the atmosphere?

Select one:

- ☒ a. It's heated from below, so is much less stable than the atmosphere ❌
- ☐ b. The coriolis effect is stronger in the ocean
- ☐ c. The ocean is heated from above, making it unstable
- ☐ d. It is heated from above, so is much more stable than the atmosphere

Your answer is incorrect.

The correct answer is: It is heated from above, so is much more stable than the atmosphere

Which factors explain why it is so wet near the equator?

Select one:

- ☐ a. The high intensity of incoming solar radiation.
- ☐ b. The warm temperatures mean the air is very humid
- ☐ c. The Walker circulation pushes moist air into these regions
- ☐ d. Large scale convection as part of the Hadley cells occurs here
- ☐ e. The thermohaline circulation drives warm water towards the equator
- ☐ f. All of the above
- ☐ g. a, c and e only
- ☒ h. a, b and d only ✔️

An increase in the mean (location) of the distribution of temperatures at a particular location, without a change in the variance (scale) or symmetry (shape) of the distribution, implies that:

Select one:

- ☒ a. there would be an increase in hot extremes and a decrease in cold extremes ✓
- ☐ b. there would be a decrease in hot extremes and a decrease in cold extremes
- ☐ c. there would be a decrease in hot extremes and an increase in cold extremes
- ☐ d. there would be an increase in hot extremes and an increase in cold extremes

Your answer is correct.

The correct answer is: there would be an increase in hot extremes and a decrease in cold extremes

Why do we find the world's deserts around 30° latitude?

Select one:

- ☐ a. The Coriolis effect is strong there
- ☐ b. The polar cells descend there, taking all the water from the air
- ☐ c. The ITCZ only reaches that far once a year
- ☒ d. The descending branch of the Hadley and Ferrel cells are dry and cool ✓

The longest continuous observed records of surface air temperature at any location are about:

Select one:

- ☒ a. 30 years long ❌
- ☐ b. 300 years long
- ☐ c. 2000 years long
- ☐ d. 100 years long
- ☐ e. 150 years long

The correct answer is: 300 years long

What is an inhomogeneity in a climate variable record?

Select one:

- ☒ a. A feature in the measurement record that is not a feature of the true climate ✔️
- ☐ b. A change in the variable's trend
- ☐ c. A significant step change in the variable
- ☐ d. A change in the variable's variability

During an El Niño event, Australia is typically ...

Select one:

- ☐ a. Unaffected
- ☐ b. Cooler and wetter
- ☐ c. More prone to flooding in the north of the country
- ☒ d. Warmer and dryer ✓

Your answer is correct.

The correct answer is: Warmer and dryer

The day/night temperature difference is much larger on land than the ocean because:

Select one:

- ☐ people live on the land surface
- ☐ the land surface is not smooth like the ocean
- ☐ the ocean has a higher heat capacity
- ☐ the land has a higher albedo
- ☐ the land has a higher heat capacity
- ☒ the ocean absorbs more energy ✗

How has the frequency of extreme rainfall changed over the 20th century?

Select one:

- ☒ a. It is likely that there have been increases in the number of heavy precipitation events in more regions than there have been decreases
- ☐ b. It is likely that there has been a decrease in the number of heavy precipitation events at the global scale
- ☐ c. It is likely that there has been an increase in the number of heavy precipitation events at the global scale
- ☐ d. There is low confidence that there are any robust long-term changes in the number of heavy precipitation events

Your answer is correct.

The correct answer is: It is likely that there have been increases in the number of heavy precipitation events in more regions than there have been decreases

Oceanic water best absorbs CO₂ when:

Select one:

- ☐ the seawater is cold and atmospheric CO₂ concentration is low (relative to oceanic CO₂)
- ☒ the seawater is cold and atmospheric CO₂ concentration is high (relative to oceanic CO₂) ✓
- ☐ the seawater is warm and atmospheric CO₂ concentration is high (relative to oceanic CO₂)
- ☐ the seawater is warm and atmospheric CO₂ concentration is low (relative to oceanic CO₂)

How do we know that the increasing CO₂ in the atmosphere is due to human activities? Choose the best answer.

Select one:

- ☐ a. Decreasing amount of Carbon 14 relative to Carbon 12 in the atmosphere
- ☐ b. The amount we've burned is roughly the amount we've added to the atmosphere
- ☐ c. Humans have been burning fossil fuels for a long time
- ☐ d. Humans now populate all of the continents on Earth
- ☒ e. Atmospheric Oxygen levels have been decreasing ❌
- ☐ f. All of these reasons
- ☐ g. a, b and e only
- ☐ h. c and d only

Your answer is incorrect.

The correct answer is: a, b and e only

An inhomogeneity in a variable's record might be caused by (choose the best answer):

Select one:

- ☐ a. Periodic instrument failure
- ☐ b. A change in the person taking the measurements
- ☐ c. A change in the time of day that measurements were taken
- ☐ d. Changes in the environment where the measurements were taken
- ☐ e. A bias in the instruments recording the variable
- ☐ f. A change in the instruments recording the variable
- ☒ g. All of the above ❌
- ☐ h. b, c, d and f only
- ☐ i. a and e only

Your answer is incorrect.

The correct answer is: b, c, d and f only

Warmer temperatures leading to melting land ice are:

Select one:

- ☒ a. Likely to further increase temperatures via the ice-albedo positive feedback ✓
- ☐ b. Likely to further increase temperatures via the ice-albedo negative feedback
- ☐ c. Likely to decrease temperatures via the ice-albedo positive feedback
- ☐ d. Likely to decrease temperatures via the ice-albedo negative feedback

Your answer is correct.

The correct answer is: Likely to further increase temperatures via the ice-albedo positive feedback

Why might we observe no change in the frequency of a climate extreme? Choose the best answer.

Select one:

- ☐ a. There has been no change in the frequency of the extreme
- ☐ b. The variability in the frequency of the extreme is too high for a real trend to be detected
- ☐ c. There is insufficient data available to assess changes in the extreme over a long period
- ☒ d. All of the above ✓
- ☐ e. b and c only
- ☐ f. a and b only

Why are anthropogenic greenhouse gas emissions a problem? Choose the best answer.

Select one:

- ☐ a. They enhance the natural greenhouse effect.
- ☐ b. The speed at which we emit them is too rapid for the climate system to absorb them into other sinks.
- ☐ c. It may take centuries for the carbon cycle to reach a new stable equilibrium.
- ☐ d. They reduce the amount of energy reflected from the Earth's surface
- ☒ e. All of the above ❌
- ☐ f. a, b and c only
- ☐ g. a and d only

Your answer is incorrect.

The correct answer is: a, b and c only

During an El Niño event, sea surface temperatures in the eastern Pacific ocean ...

Select one:

- ☒ a. Get warmer than usual ✔️
- ☐ b. Get cooler than usual
- ☐ c. Don't change

What are the three types of error in global climate data?

Select one:

- ☐ a. Station error, bias error, and computer error
- ☒ b. Station error, sampling error, and bias error ✓
- ☐ c. Statistical error, sampling error, and computer error
- ☐ d. Bias error, sampling error, and statistical error

Your answer is correct.

The correct answer is: Station error, sampling error, and bias error

From the start of the Industrial Revolution to 2016, atmospheric CO₂ concentrations have:

Select one:

- ☐ a. risen from about 80 ppm to 90 ppm
- ☐ b. declined from about 400 ppt to 280 ppt
- ☐ c. risen from about 280 ppt to 400 ppt
- ☐ d. declined from about 400 ppm to 280 ppm
- ☒ e. risen from about 280 ppm to 400 ppm ✓

Which of the following would increase the anthropogenic greenhouse effect the most?

Select one:

- ☒ a. Emitting 1000 tons of carbon dioxide ❌
- ☐ b. Emitting 5 tons of Nitrous oxide
- ☐ c. Emitting 50 tons of oxygen
- ☐ d. Emitting 100 tons of methane
- ☐ e. Emitting 2000 tons of nitrogen

The correct answer is: Emitting 100 tons of methane

The 'greenhouse effect' was first identified around:

Select one:

- ☐ 1990
- ☒ 1850 ✔️
- ☐ 1970
- ☐ 1960

Atmospheric circulation on large scales is fundamentally driven by:

Select one:

- ☐ a. El Nino Southern Oscillation (ENSO) cycles
- ☒ b. The Coriolis effect ❌
- ☐ c. Convection from mid-latitude deserts
- ☐ d. Thermohaline circulation
- ☐ e. Hadley and Polar cells
- ☐ f. Equator-pole energy imbalance

Your answer is incorrect.

The correct answer is: Equator-pole energy imbalance

How many circulation cells would Earth have if it did not spin?

Select one:

- ☐ a. None
- ☐ b. Three
- ☒ c. Two ✔️
- ☐ d. One
- ☐ e. Six

Which one of the following statements relate to climate as opposed to weather?

Select one:

- ☒ We've now had five days in a row above 40C ❌
- ☐ The first two months of 2013 were very wet
- ☐ June is the wettest month in Sydney on average
- ☐ Last winter was the coldest I can remember

The correct answer is:

June is the wettest month in Sydney on average

The Coriolis effect is a result of:

Select one:

- ☐ a. Ocean-atmosphere interaction
- ☐ b. Atmospheric circulation patterns
- ☐ c. Earth's orbital distance from the sun
- ☐ d. Ocean circulation patterns
- ☒ e. Earth's rotation ✔️

A "bottom-up" extreme event is:

Select one:

- ☐ a. one that has an unusually large impact on society
- ☐ b. one that no-one can remember ever experiencing before
- ☐ c. one that involves unusually large insured losses
- ☒ d. one that is defined as statistically rare in terms of climate variables ❌

The correct answer is: one that has an unusually large impact on society

The thermohaline circulation is driven by:

Select one:

- ☐ a. ocean gyre currents hitting continental boundaries
- ☒ b. changes in the salinity and temperature of sea water ✔️
- ☐ c. El Niño - Southern Oscillation
- ☐ d. the East Australia Current (EAC)
- ☐ e. the oceanic solubility carbon pump

What is the net effect of *anthropogenic* aerosols (e.g. sulfate aerosols from fossil fuel combustion) on the climate?

Select one:

- ☒ a. Overall they warm the Earth by enhancing the natural greenhouse effect ❌
- ☐ b. Their warming and cooling effects cancel each other out, so there is no net affect.
- ☐ c. Overall they warm the Earth by trapping reflected solar radiation
- ☐ d. Overall they cool the Earth by reducing the natural greenhouse effect
- ☐ e. Overall they cool the Earth by reflecting solar radiation

Your answer is incorrect.

The correct answer is: Overall they cool the Earth by reflecting solar radiation

Which of the following describes the SAM?

Select one:

- ☐ a. A multi-decadal oscillation of sea surface temperatures in the Northern Pacific
- ☐ b. The annual peak rainfall period in the southern hemisphere
- ☐ c. The direction and strength of the Walker circulation
- ☒ d. The strength and position of Southern Hemisphere westerly winds ✔️
- ☐ e. The variability of sea surface temperatures in the North Atlantic ocean

Why do scientists homogenise raw weather observations?

Select one:

- ☐ a. To hide global cooling
- ☐ b. To reduce the **variability** in recorded data and make trends clearer
- ☐ c. To make temperature trends align better with model projections
- ☐ d. To account for the uneven distribution of observations across space and time
- ☒ e. To account for false trends caused by changes in the recorded data unrelated to climate ✓

Your answer is correct.

The correct answer is: To account for false trends caused by changes in the recorded data unrelated to climate

The East Australian Current transports:

Select one:

- ☒ warm water south ✓
- ☐ cold water south
- ☐ warm water north
- ☐ cold water north
- ☐ Nemo and his friends north

A "top-down" extreme event is:

Select one:

- ☐ a. one that has an unusually large impact on society
- ☐ b. one that no-one can remember ever experiencing before
- ☒ c. one that is defined as statistically rare in terms of climate variables ✓
- ☐ d. one that involves unusually large insured losses

The correct answer is: one that is defined as statistically rare in terms of climate variables

Suppose that we have a parcel of air that had relative humidity below 100%, and the amount of water vapour within it is increased (for example, if the [parcel](#) of air were over a lake). Which one of the following statements is true?

Select one:

- ☒ a. If the temperature of the air does not change, the relative humidity will increase. ✓
- ☐ b. If the temperature decreases, the relative humidity will decrease.
- ☐ c. If the temperature of the air does not change, the relative humidity will decrease.
- ☐ d. The relative humidity will increase regardless of any temperature changes.
- ☐ e. None of the above

Globally, which sector emits the most carbon dioxide?

Select one:

- ☐ a. Forestry
- ☐ b. Transport
- ☐ c. Agriculture
- ☒ d. Energy generation ✓
- ☐ e. Industry

Your answer is correct.

The correct answer is: Energy generation

El Niño - Southern Oscillation (choose the best answer):

Select one:

- ☒ a. is a coupled ocean-atmosphere phenomenon ✓
- ☐ b. involves only changes in ocean temperature
- ☐ c. involves only the atmosphere - specifically changes in rainfall
- ☐ d. is not related to the ocean or the atmosphere
- ☐ e. occurs because of the seasons
- ☐ f. c and e only

Subtropical gyres are:

Select one:

- ☒ a. driven by trade winds and mid-latitude westerly winds ✓
- ☐ b. areas of low atmospheric pressure that lead to convection
- ☐ c. caused by ocean salinity gradients
- ☐ d. driven by deep water formation
- ☐ e. caused by El Niño - Southern Oscillation (ENSO)

Your answer is correct.

The correct answer is: driven by trade winds and mid-latitude westerly winds

A probability density function shows:

Select one:

- ☒ a. the relative likelihood of different values of a variable ✓
- ☐ b. whether the mean of a variable is likely to change
- ☐ c. whether the skewness of a variable is likely to change
- ☐ d. whether the variance of a variable is likely to change

What is the defining characteristic of an unstable climate state?

Select one:

- ☒ a. It is dominated by positive feedbacks ✓
- ☐ b. It is caused by increasing levels of atmospheric CO₂
- ☐ c. It is dominated by negative feedbacks
- ☐ d. It rapidly becomes much warmer

Your answer is correct.

The correct answer is: It is dominated by positive feedbacks

Which of the following carbon sinks contains the most carbon?

Select one:

- ☐ a. Soils
- ☐ b. Forests
- ☐ c. The atmosphere
- ☒ d. The ocean ✓

Suppose you had two identical jars, one containing air at 15°C and one containing air at 26°C. If both had relative humidity of 100%:

Select one:

- ☐ Both jars would contain the same amount of water vapour
- ☐ The first jar would contain more water vapour
- ☐ More information is required to know which jar has more water vapour
- ☒ The second jar would contain more water vapour ✓

The correct answer is:

The second jar would contain more water vapour

Which source or sink of CO₂ dominates the annual atmospheric CO₂ cycle?

Select one:

- ☐ a. Industrial emissions in the Southern Hemisphere
- ☐ b. Zooplankton
- ☐ c. The solubility pump
- ☐ d. Industrial emissions in the Northern Hemisphere
- ☐ e. Emissions from volcanoes
- ☐ f. Southern Hemisphere plants
- ☒ g. Northern Hemisphere plants ✓